

Dimensionality Reduction

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Introduction

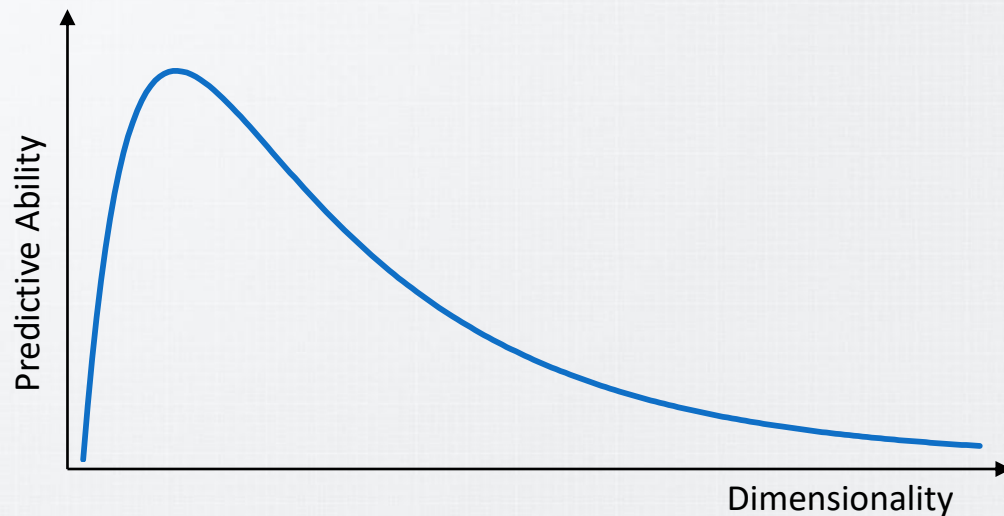
Dimensionality Reduction

- In some scenarios, the data comprises a large number of features (dimensionality)
 - $X \in \mathbb{R}^{n \times 10^6}$
- Main issues involving data dimensionality
 - Computational cost
 - Theoretical problems

The Curse of Dimensionality

Dimensionality Reduction

- For a fixed sample size, increasing the number of features beyond a certain value degrades the predictive ability of the model rather than improving it



The Sample Size and Related Problems

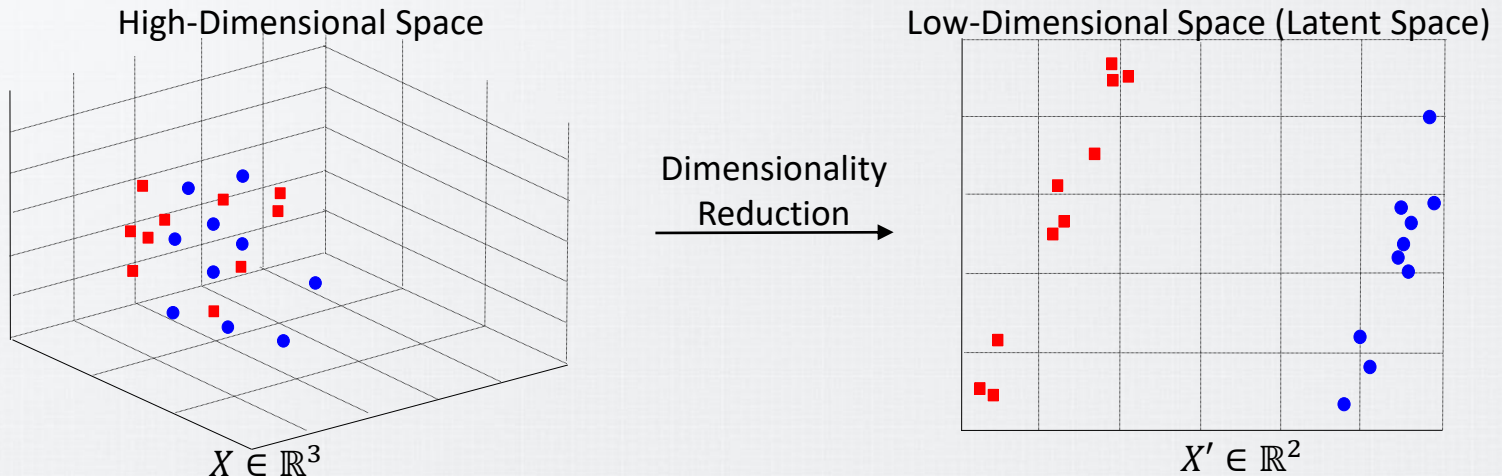
Dimensionality Reduction

- Ordinary least squares is not applicable when $m > n$
 - The sample covariance matrix ($X^T X$) is singular; hence, it is not invertible
- *“The same authors then introduced TabPFNv2 [...] significantly outperforms tree-based and neural network competitors on small-to-medium datasets with up to 10K samples and **500 features**”. (Qu et al., 2025)*

Problem Definition

Dimensionality Reduction

- Dimensionality reduction aims to construct a new set of features by **combining the original ones**
- Given a feature space $\mathbb{R}^m$, this family of techniques seeks a mapping to a lower-dimensional space (**latent space**) $\mathbb{R}^c$, $c \ll m$, that preserves key properties of the original representation



Popular Techniques

Dimensionality Reduction

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Partial Least Squares (PLS)
- Canonical Correlation Analysis (CCA)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

Singular Value Decomposition (SVD)

Problem Definition

Singular Value Decomposition

- Assume a matrix $A \in \mathbb{R}^{n \times m}$ and $r \in \mathbb{N}$
- Let r be the rank of A
 - Number of linearly independent columns of A
 - $\begin{cases} \text{full rank, if } r = \min(n, m) \\ \text{low rank, if } r \ll \min(m, n) \end{cases}$
- A rank- r approximation problem requires finding a matrix A' that minimizes
 - $\|A - A'\|_2$
 - Satisfying $\text{rank}(A') \leq r$
- Eckart–Young–Mirsky theorem provides an optimal solution of this problem using Singular Value Decomposition (SVD)

Overview

Singular Value Decomposition

- The SVD decomposes a matrix $A \in \mathbb{R}^{n \times m}$ such that $A = U\Sigma V^T$
 - U and V have orthonormal columns representing the **singular vectors** of A
 - $\Sigma \in \mathbb{R}^{n \times m}$ is a diagonal matrix containing the **singular values** of A
 - Singular vectors, in descending order, reveal the most important axes of transformation of the matrix
- $U \in \mathbb{R}^{n \times n}$
 - Orthonormal basis of $\mathbb{R}^n$. $U^T U = I$
- $V \in \mathbb{R}^{m \times m}$
 - Orthonormal basis of $\mathbb{R}^m$. $V^T V = I$
- U and V^T represent rotations that matrix A performs while Σ represents scaling along those axes

$$\begin{array}{c} m \\ \boxed{A} \\ n \end{array} = \begin{array}{c} n \\ \boxed{U} \\ n \end{array} \begin{array}{c} m \\ \boxed{\Sigma} \\ n \end{array} \begin{array}{c} m \\ \boxed{V^T} \\ m \end{array}$$

Practical Example

Singular Value Decomposition

- Full matrices
 - $U \in \mathbb{R}^{n \times n}$
 - $\Sigma \in \mathbb{R}^{n \times m}$
 - $V^T \in \mathbb{R}^{m \times m}$
- Reduced matrices (thin SVD)
 - $r = \min(n, m)$
 - $U \in \mathbb{R}^{n \times r}$
 - $\Sigma \in \mathbb{R}^{n \times m}$
 - $V^T \in \mathbb{R}^{r \times m}$

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Principal Component Analysis (PCA)

Preliminaries

Principal Component Analysis

- Covariance ($\text{Cov}(\cdot)$) of $X \in \mathbb{R}^{n \times m}$ (zero-mean) is an $m \times m$ matrix and indicates the relationship between two random variables
 - $\text{Cov}(X) = \frac{1}{n} X^T X \Rightarrow \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^T (\mathbf{x}_i - \boldsymbol{\mu})$
 - $\boldsymbol{\mu} \in \mathbb{R}^m$ is the mean sample

$$\begin{bmatrix} E[(X^1 - \mu^1)^2] & \dots & E[(X^1 - \mu^1)(X^m - \mu^m)] \\ & \ddots & \\ E[(X^1 - \mu^1)(X^m - \mu^m)] & \dots & E[(X^m - \mu^m)^2] \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & \dots & c_{1m} \\ & \ddots & \\ c_{1m} & \dots & \sigma_m^2 \end{bmatrix}$$

Preliminaries

Principal Component Analysis

- The covariance matrix is **symmetric** and tells us something about the spread of the data
 - The diagonal entries correspond to the variances of the individual variables
 - Its top **singular vectors** are the directions of **maximal variance** in the data and the associated variances are proportional to the squared singular values
- Rayleigh-Ritz Theorem or Maximum Principle for Symmetric Matrices
 - $z(w) = \frac{w^T A w}{w^T w}$, $w \neq 0$ and A is symmetric
 - The maximum and minimum values of z are $\lambda_1(A)$ and $\lambda_k(A)$, respectively, assuming $\lambda_1 \geq \lambda_2 \dots, \lambda_k$

Definitions

Principal Component Analysis

- Principal components are vectors that define a new coordinate system in which the **first axis** goes in the direction of the **highest variance** in the data
- The second axis is orthogonal to the first one and goes in the direction of the second highest variance in the data
- Each new axis is **orthogonal** to all previously defined axes and aligns with the direction of the next highest **remaining variance**
 - Subject to the constraint of zero correlation (uncorrelated) with the preceding components

Problem Definition

Principal Component Analysis

- Find a set of orthogonal directions that capture the maximum variance of the data
- $\max_{\|w\|_2=1} \text{Var}(w^T X)$
 - $\|w\|_2 = 1$ prevents arbitrary scaling
 - Here $X \in \mathbb{R}^{m \times n}$ - samples as column vector
- $$\begin{aligned} \text{Var}(w^T X) &= E[(w^T X - w^T \mu)^2] = E[(w^T X - w^T \mu)(w^T X - w^T \mu)] \\ &= E[w^T (X - \mu)(X - \mu)^T w] = w^T E[(X - \mu)(X - \mu)^T] w \\ &= \mathbf{w}^T \mathbf{Cov}(\mathbf{X}) \mathbf{w}. \end{aligned}$$
- Therefore, $\max_{\|w\|_2=1} \text{Var}(w^T X) = \max_{\|w\|_2=1} \mathbf{w}^T \mathbf{Cov}(\mathbf{X}) \mathbf{w}$

PCA with SVD

Principal Component Analysis

- PCA is theoretically an eigenvalue problem, but most frameworks solve it through the SVD because it is more numerically stable and **avoids explicitly computing the covariance matrix**
- $X = U\Sigma V^T$
 - U, Σ and V come from the SVD decomposition
- $X^T X = (U\Sigma V^T)^T (U\Sigma V^T) = V\Sigma^T U^T U\Sigma V^T$
- As $U^T U = I$, we have $X^T X = V\Sigma^T \Sigma V^T$
- $\Sigma^T \Sigma = \Sigma^2$
 - Therefore, $X^T X = V\Sigma^2 V^T$

Important note

The eigenvectors of $A^T A$ are the right-singular vectors V of A

The eigenvalues of $A^T A$ are the squared singular values of Σ ($\lambda_i = \sigma_i^2$)

(see Equations 4.71 to 4.75 of MATHEMATICS FOR MACHINE LEARNING)

PCA with SVD

Principal Component Analysis

- V^T
 - **Projection Matrix**
 - Each row is a principal component
- $U\Sigma$
 - Data projected onto the principal components

$$\begin{matrix} m \\ n \end{matrix} \boxed{X} = \begin{matrix} n \\ n \end{matrix} \boxed{U} \begin{matrix} m \\ n \end{matrix} \boxed{\Sigma} \begin{matrix} m \\ m \end{matrix} \boxed{V^T}$$

- We select how many dimensions ($\leq m$) we want in the new space
 - Selecting the first k rows of V^T
- Alternatively, we can also solve PCA by computing eigenvalues and eigenvectors from the covariance matrix

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Proportion of Variance Explained

Principal Component Analysis

- The proportion of variance explained by the first k components is the ratio between the sum of the k **largest eigenvalues** and the total variance

Property	Summary
Variance by component i	σ_i^2
Variance explained by component i	$\frac{\sigma_i^2}{\sum \sigma_i^2}$

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Linear Discriminant Analysis (LDA)

Introduction

Linear Discriminant Analysis

- A key limitation of PCA is that it ignores class separability, as it does not incorporate the class labels associated with the feature vectors
 - $U\Sigma V^T = SVD(X)$
 - **Observe that PCA does not take into account Y**
- PCA performs a coordinate rotation that aligns the new axes (V^T) with the directions of highest variance

Introduction

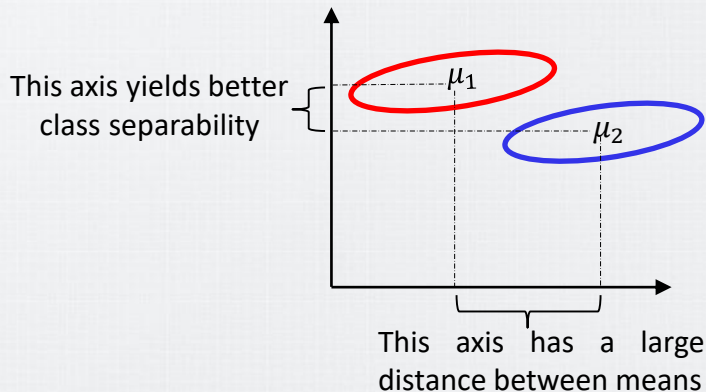
Linear Discriminant Analysis

- Linear Discriminant Analysis (LDA) is a supervised method for dimensionality reduction for classification problems: it employs both **X** and **Y** in the process to generate a new latent space
- LDA seeks to reduce dimensionality while preserving as much of the class discriminatory information as possible

Problem Definition

Linear Discriminant Analysis

- In order to find a good latent space, we must first define a criterion that quantifies class separation
- Consider the distance between the **projected means** as the objective function $J(\cdot)$
 - $J(w) = w^T(\mu_1 - \mu_2)$
 - Unfortunately, the distance between projected means does not provide a reliable measure, as it fails to account for within-class variability (standard deviation)



Problem Definition

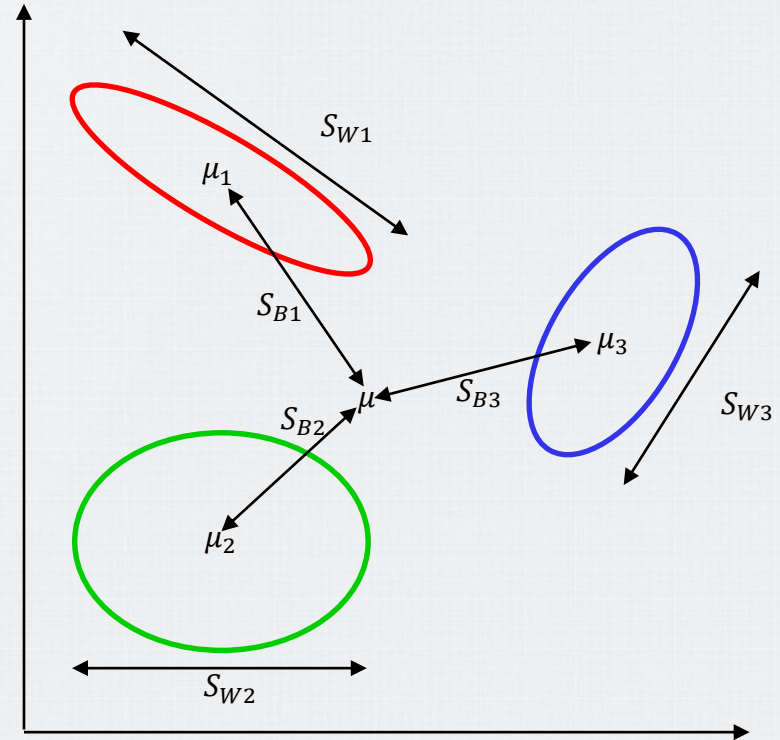
Linear Discriminant Analysis

- Fisher suggested maximizing the difference between class means while normalizing it by a measure of within-class scatter
- Let $S_W = \sum_{i=1}^C S_i$ be the within-class scatter
 - $S_i = \sum (x - \mu_i)^T (x - \mu_i)$, where x and μ_i are the samples and mean of class i
- Let $S_B = \sum_{i=1}^C n_i (\mu_i - \mu)^T (\mu_i - \mu)$ be the between-class scatter
 - n_i stands for the number of samples of class i . This weighting accounts for class size and yields a statistically consistent estimate when classes are imbalanced
- $J(W) = \frac{\det(W^T S_B W)}{\det(W^T S_W W)}$
 - LDA seeks the projection matrix W^* that maximizes this ratio

Problem Definition

Linear Discriminant Analysis

- $J(W) = \frac{\det(W^T S_B W)}{\det(W^T S_W W)}$
 - S_W within-class scatter
 - S_B between-class scatter



Algorithm

Linear Discriminant Analysis

- The optimal projection matrix $W^*(w_1, w_2, \dots w_{classes-1})$ consists of the eigenvectors associated with the largest eigenvalues of the corresponding generalized eigenvalue problem
 - $W^*[w_1^*, w_2^*, \dots w_{classes-1}^*] = \underset{W}{argmax} \frac{\det(W^T S_B W)}{\det(W^T S_W W)}$
- The projections with maximum class separability information are the eigenvectors corresponding to the largest eigenvalues of $S_W^{-1} S_B$
 - S_W should be invertible

Other Techniques

Partial Least Squares

Other Techniques

- Partial Least Squares (PLS) aims at maximizing the covariance between the independent (X) and dependent (Y) variables
- $w_i = \text{maximize}(Cov(Xw, Y)), s. t. \|w\|_2 = 1$
 - The exact solution for this criterion is $w_i = \frac{X^T Y}{\|X^T Y\|_2}$

NIPALS for binary PLS($X, Y \in \mathbb{R}^{n \times 1}$)

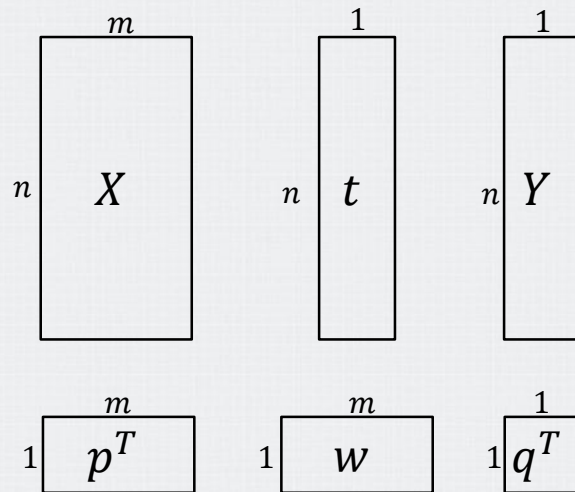
For $i = 1$ to c do

$$w_i = \frac{X^T Y}{\|X^T Y\|_2}$$

$$t_i = X w_i$$

$$q_i = \frac{Y^T t_i}{\|Y^T t_i\|_2}, p_i = \frac{X^T t_i}{\|X^T t_i\|_2}$$

$$X = X - t_i p_i^T, Y = Y - t_i q_i^T$$



Random Projections

Other Techniques

- Random projection involves projecting the original data onto a subspace using a **random matrix** $R \in \mathbb{R}^{m \times c}$ (Bingham et al., 2001)
 - The rows have unit lengths
- The key idea of random mapping arises from the Johnson-Lindenstrauss lemma
 - if points in a vector space are projected onto a randomly selected subspace of suitably high dimension, then the distances between the points are approximately preserved

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Comparison between Techniques

Other Techniques

- Each technique focuses on different properties to find the projection matrix

Technique	Property (Summary)
PCA	Maximizing variance
LDA	Maximizing the ratio of between-class scatter to within-class scatter
PLS	Maximizing the covariance between latent variables of X and Y
Random	Simplicity

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Learning and Testing Phases

Other Techniques

- Learn a projection matrix $W \in \mathbb{R}^{m \times c}$
 - c is a parameter and stands for the number of axes (components) in the new latent space
 - In this phase, we use X^{Train}
- Project both X^{Train} and X^{Test} onto the latent space
 - After the projection, we can train a classifier (or any predictive model) in the latent space

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Incremental Techniques

Motivation

Incremental Techniques

- Traditional dimensionality reduction techniques may be unfeasible for large datasets since all the data need to be available in advance
 - Memory constraints (e.g., consider $X \in \mathbb{R}^{10^6 \times 10^4}$)
- Incremental dimensionality reduction methods find the projections using a single data sample (or batch of data) at a time
 - Keep some properties of the traditional dimensionality reduction methods
- Useful for streaming applications
 - The task continuously generates data

For x, y in X, Y

update $\mu \triangleleft$ Typical process

update W using x or (x, y)

Popular techniques

Incremental Techniques

- Incremental PCA (Weng et al., 2003)
- Incremental LDA (Lu et al., 2012)
- Incremental PLS (Zeng et al., 2014; Arora et al., 2016)
- Some examples available [here](#)

Weng et al. *Candid covariance-free incremental principal component analysis*. Transactions on Pattern Analysis and Machine Intelligence, 2003

Lu et al. *Incremental learning of complete linear discriminant analysis for face recognition*. Knowledge-Based Systems, 2012

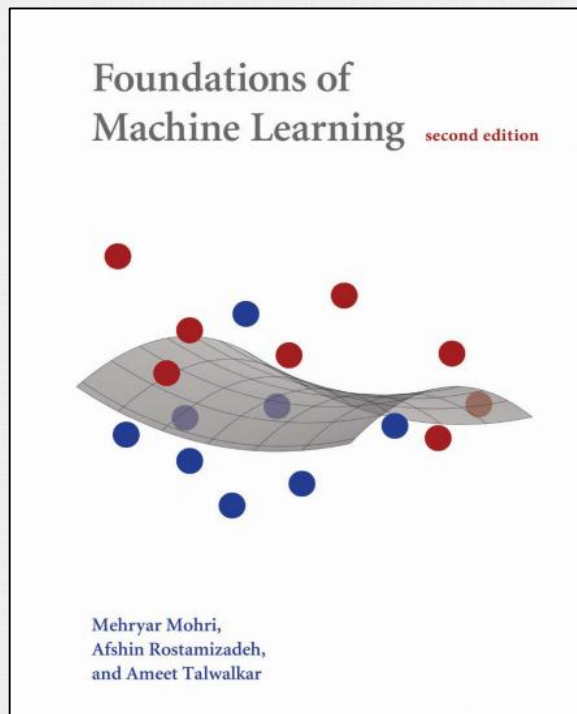
Zeng et al. *Incremental partial least squares analysis of big streaming data*. Pattern Recognition, 2014

Arora et al. *Stochastic optimization for multiview representation learning using partial least squares*. In ICML, 2016

Bibliography

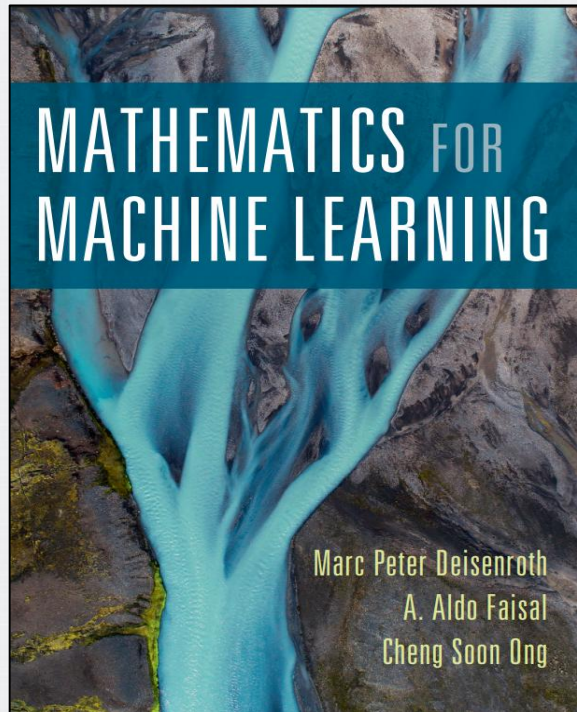
Bibliography

- Foundations of Machine Learning
 - Chapter 15
 - 15.1 Principal component analysis



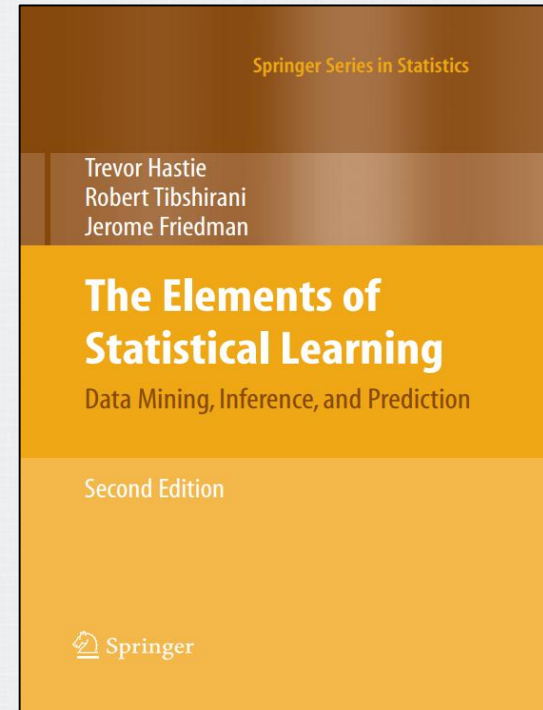
Bibliography

- MATHEMATICS FOR MACHINE LEARNING
 - Chapter 4
 - 4.5 Singular Value Decomposition
 - Observe Equations 4.71 to 4.75
 - Chapter 6
 - 6.4.1 Means and Covariances



Bibliography

- The Elements of Statistical Learning
 - Chapter 4
 - 4.3.2 Computations for LDA
 - 4.3.3 Reduced-Rank Linear Discriminant Analysis



Bibliography

- Matrix Computations
 - Chapter 8
 - 8.2.3 Rayleigh Quotient Iteration

